

## DESIGN & PROGRAM

Developer: NASA Jet Propulsion Laboratory (JPL)  
Program: CADRE – Technology Demonstration  
Type: Lunar Surface Exploration Rover  
Role: Autonomous Science / Reconnaissance  
Status: Concept / Development (2019–Present)

## SYSTEM OVERVIEW

CADRE is a small, highly autonomous rover designed to explore the lunar surface, create 3D maps of the subsurface, and demonstrate cooperative multi-robot operations with minimal human intervention.

- ✓ High autonomy & fault tolerance
- ✓ 3D subsurface characterization
- ✓ Cooperative multi-rover operations
- ✓ Scalable for future lunar missions
- ✓ Low mass, low power, cost-effective

## KEY SPECIFICATIONS (EST.)

|                       |                                   |
|-----------------------|-----------------------------------|
| 📏 Dimensions (L×W×H): | ~1.2 m × 0.9 m × 1.1 m            |
| 📊 Mass:               | ~50–70 kg                         |
| 📦 Payload Capacity:   | ~20–30 kg                         |
| 🚀 Max Speed:          | ~0.5 m/s                          |
| 🕒 Endurance:          | >14 Earth days                    |
| ⚡ Power:              | ~150–250 W (avg)                  |
| 📶 Comm Range:         | Line-of-sight to orbiter / lander |
| 🌡️ Operating Temp:    | -180°C to +120°C                  |
| 🚶 Mobility:           | 0–30° slope, 0.3 m obstacles      |

# CADRE-STYLE AUTONOMOUS ROVER

## SYSTEMS ENGINEERING ANALYSIS

COOPERATIVE AUTONOMOUS DISTRIBUTED ROBOTIC EXPLORER (CADRE) CONCEPT

### MASTED SENSORS

NavCams, payload sensors, laser altimeter, lighting for low-angle illumination

### COMMUNICATIONS

UHF / X-band radio for relay via orbiter or lander

### SOLAR PANELS

Deployable, high-efficiency solar arrays for power generation

### AI / AUTONOMY COMPUTER

Onboard processing for perception, planning, and decision making

### SUSPENSION & MOBILITY

Rocker-bogie suspension for traversing rough terrain and maintaining stability

### WHEELS

Lightweight aluminum wheels with grousers for traction on regolith

### POWER SYSTEM

Battery pack for survival during lunar night and high-power operations

### PAYLOAD BAY

Modular payloads for subsurface radar, neutron spectrometer, cameras, and tools

### GROUND PENETRATING RADAR (GPR)

3D subsurface mapping to ~10 m depth (science variant)

## MISSION CONTEXT

Designed to operate on the lunar surface as part of Artemis precursor missions. Demonstrates cooperative autonomy, 3D subsurface mapping, and scalable multi-rover operations.

Environment: Lunar Surface

Gravity: 1/6 Earth Atmosphere: None

## MISSION OPERATIONS (TYPICAL)

### 1 DEPLOYMENT

Delivered by lander and deployed to surface

### 2 SYSTEM CHECKOUT

Autonomous health check, sensor calibration, environment assessment

### 3 AUTONOMOUS TRAVERSE

Navigate to waypoints, avoid hazards, maintain situational awareness

### 4 DATA COLLECTION

Acquire surface imagery, GPR subsurface scan, environmental data

### 5 COOPERATIVE OPS

Share data with other rovers and relay via orbiter / lander

### 6 SCIENCE RETURN

Transmit data to orbiter / Earth

### 7 EXTENDED OPERATIONS

Repeat traverse, expand coverage, adapt mission based on findings

## SYSTEM ARCHITECTURE

### SENSORS

- NavCams (stereo)
- Hazard Cameras
- Laser Altimeter
- IMU / Star Tracker
- Wheel Encoders
- Environmental Sensors

### PERCEPTION

- Terrain Mapping
- Hazard Detection
- Visual Odometry
- 3D Reconstruction
- Localization

### PLANNING & DECISION

- Path Planning
- Traversability Analysis
- Multi-Rover Coordination
- Science Targeting
- Behavior Selection

### MOBILITY

- Rocker-Bogie Suspension
- 6-Wheel Drive
- Wheel Slip Control
- Terrain Adaptation

### POWER

- Solar Power
- Li-ion Batteries
- Power Management
- Thermal Control

### COMMUNICATIONS

- UHF / X-band Radio
- High-Gain Antenna
- DTN / Delay-Tolerant Networking
- Orbiter / Lander Relay

### HEALTH MANAGEMENT & FAULT PROTECTION

- 📺 Continuous Monitoring
- 🚨 Fault Detection
- 🛡️ Safe Mode
- 🔄 Autonomous Recovery
- 🕒 Watchdog Timers
- 🛡️ Radiation Tolerant Electronics

## PAYLOAD OPTIONS (EXAMPLES)



GROUND PENETRATING RADAR (GPR)



NEUTRON SPECTROMETER



MULTISPECTRAL CAMERA



MICRO-DRILL / SAMPLING TOOL

## AUTONOMY & PERFORMANCE

- ✓ High autonomy: operates with minimal human intervention
- ✓ Real-time hazard avoidance and route replanning
- ✓ Cooperative multi-rover mapping and data fusion
- ✓ Subsurface mapping to support science and resource identification
- ✓ Designed for reliability in extreme lunar environment

## MISSION OBJECTIVES

- 📊 Map lunar subsurface in 3D
- 👥 Demonstrate cooperative autonomy
- 🚀 Scout landing zones & future exploration sites
- 🔭 Support Artemis science & ISRU prospecting

## KEY DESIGN DRIVERS

- 📊 Low Mass & Low Cost
- 🎯 High Autonomy
- 🔄 Scalability & Modularity
- ⚙️ Robustness & Fault Tolerance

## ENGINEERING CONSIDERATIONS

- ✓ Thermal extremes
- ✓ (-180°C to +120°C)
- ✓ Lunar dust & abrasion
- ✓ Radiation environment
- ✓ Communication delays
- ✓ Power during lunar night

## RISK MITIGATION

- ✓ Redundant critical systems
- ✓ Health monitoring & safe modes
- ✓ Autonomous fault recovery
- ✓ Cooperative operations increase robustness

## REFERENCES

NASA JPL – CADRE Project: <https://www.jpl.nasa.gov/missions/cadre/>  
NASA – CADRE Overview: <https://www.nasa.gov/directorates/spacetech/cadre/>

NASA – VIPER Lunar Rover: <https://science.nasa.gov/mission/viper/>  
NASA – Artemis Program: <https://www.nasa.gov/artemis/>

Wolff, M. J., et al. (2019). "CADRE: Cooperative Autonomous Distributed Robotic Explorers." 2019 IEEE Aerospace Conference.

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